# TL;DR

- In factor level, the diffusion model performance is the same level as the resample baseline, and both are better than the Gaussian baseline.
- Diffusion model fail to capture the factor relationship.
- All methodology have hard time in stock level evaluation, and diffusion model is even worse, and the problem is more in the idiosyncratic modeling than the systematic modeling.

# Experimental Setup

Datasets: 2001 - 2025 daily returns and characteristics. 2025 are held out as OOS test set (261 samples), 2001-2024 are used for training (6262 samples).

Baselines:

- Resample: Randomly resample rows from the training dataset with replacement.
- Gaussian: Fit a multivariate Gaussian to the training data and sample from it. (factor evaluation only)

The evaluation are in factor level and stock level. For factor evaluation. The selected factor returns are `[market, growth, momentum, quality, size, value, volatility]` which are generated via OLS on each date. For stock evaluation, we only evaluate the stocks that both in the training set and test set.

```
model/regression
```

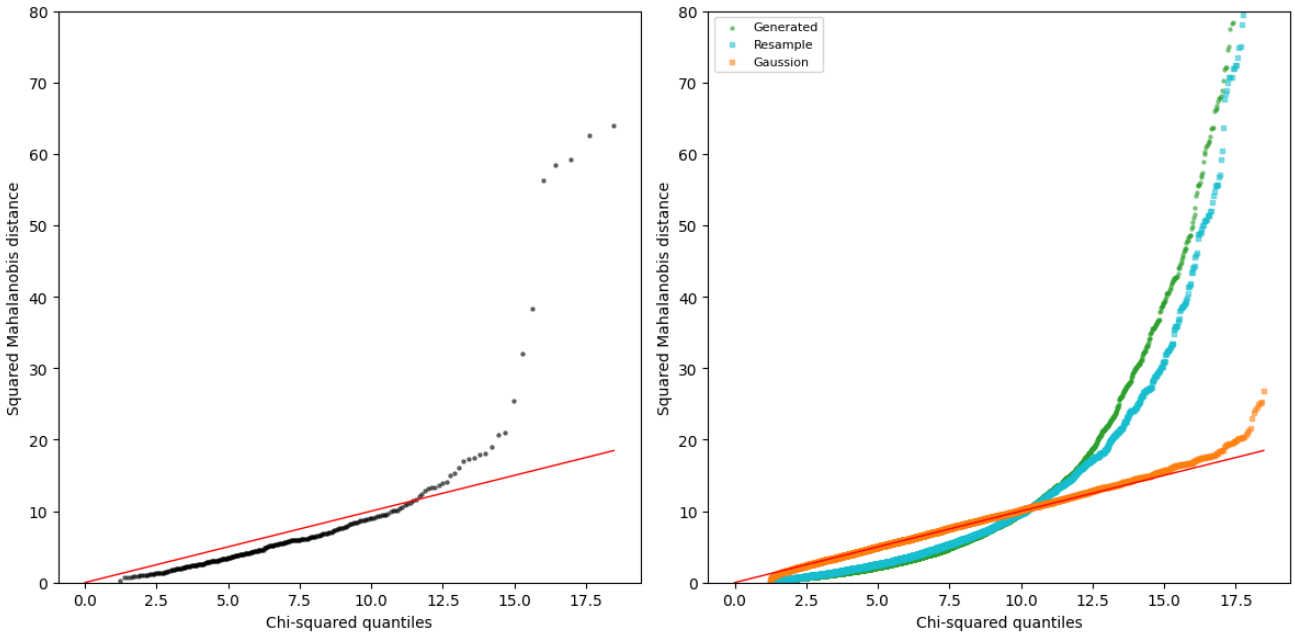
```
model/regression/samples/factor_4096.npy
```

```
OOS: (260, 7)  sampled: (4096, 7)  Gaussian: (4096, 7)
```

# Mahalanobis Check

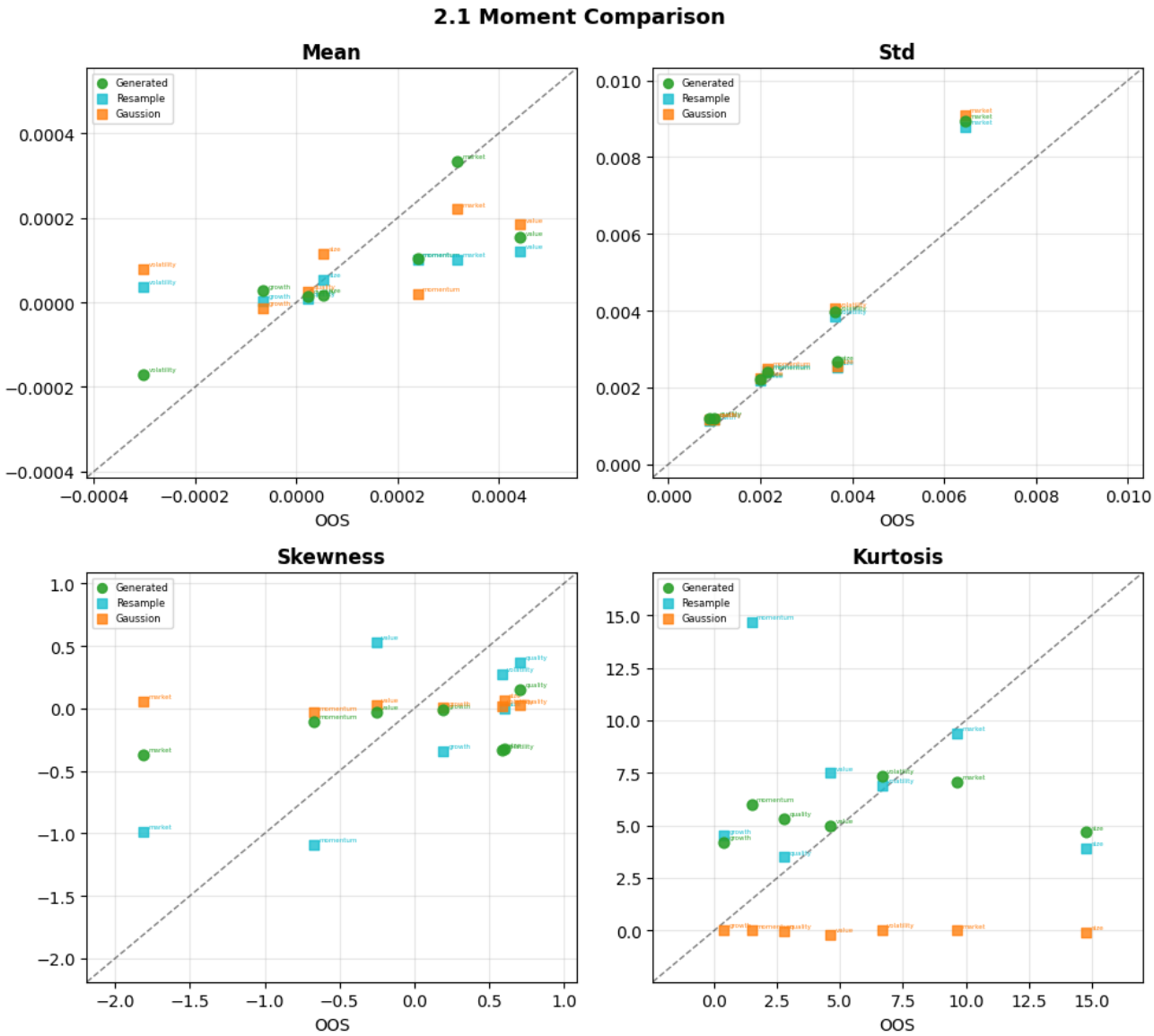
Mahalanobis distance is a multivariate measure of how far a point is from the mean of a distribution, taking into account the covariance structure. If the generated samples are from the same distribution as the OOS data, their squared Mahalanobis distances should approximately follow a chi-squared distribution with degrees of freedom equal to the number of factors.

Left graph shows the result of the OOS data, it can be seen that our factor return has fat tail. Right graph show our generated samples and the resample/multivariate baseline. And the diffusion model can capture the tail behavior while multivariate Gaussian fails to do




# Moment Comparison (factor level)

Comparing moments (mean, std, skewness, kurtosis) per factor of the generated samples and baseline against the OOS data. we use MAE (mean absolute error) to summarize the deviation of each moment across factors. The scatter plots show the per-factor moments of generated/baseline vs OOS, with a 45-degree reference line. From the illustration, all methodology can capture the mean and std well, diffusion model capture the kurtosis better and resample baseline capture the skewness better. And they are all better than the Gaussian baseline.



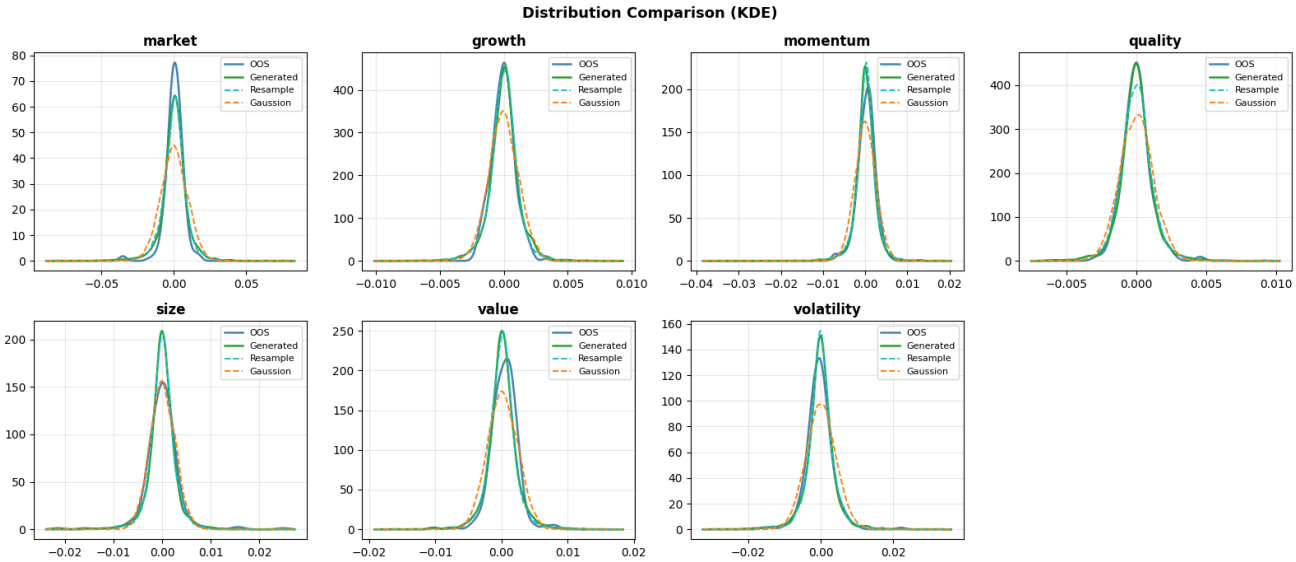

```
MAE across factors — mean (std):
Moment       Generated              Resample               Gaussion
-------------------------------------------------------------------------------
Mean         0.0001 (0.0001)        0.0002 (0.0001)        0.0002 (0.0001)
Std          0.0007 (0.0008)        0.0007 (0.0008)        0.0007 (0.0008)
Skewness     0.6919 (0.4065)        0.5458 (0.1893)        0.6811 (0.5136)
Kurtosis     3.5210 (3.0251)        4.6070 (4.9193)        5.8091 (4.7249)
```

# Distribution Comparison

We draw KDE(kernel density estimation) plots for each factor, comparing the OOS distribution with the generated samples and baselines. We also compute the Wasserstein-1 distance to measure similarity between marginal distributions.

The KDE plots show that the diffusion model (Generated) closely tracks the OOS density across all seven factors, reproducing both the sharp peak and fat tails. The Gaussian baseline consistently underestimates peak density and underestimates tails. Resample performs comparably to Generated in shape.

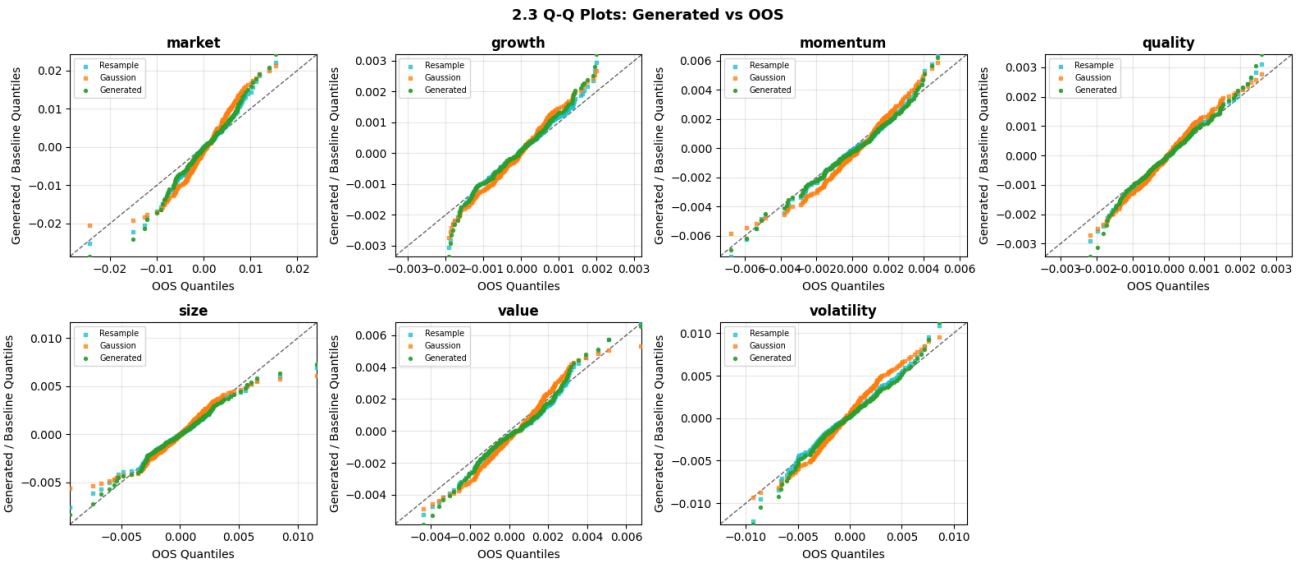
The Wasserstein-1 distances shows Generated achieves W1 comparable to or lower than Resample for all factors, while Gaussian is consistently higher — most notably for `market` (0.0032 vs 0.0018) and `volatility` (0.0009 vs 0.0004).




```
Wasserstein-1 Distance:
Factor          Generated      Resample       Gaussion
-------------------------------------------------------
market          0.0018         0.0018         0.0032
growth          0.0002         0.0002         0.0002
momentum        0.0003         0.0003         0.0005
quality         0.0001         0.0001         0.0002
size            0.0004         0.0004         0.0005
value           0.0004         0.0004         0.0005
volatility      0.0004         0.0004         0.0009
```

# QQ plot

Q-Q plots compare the quantiles of the generated samples and baselines against the OOS data for each factor. It gives another view of the factor distributional similarity, especially in the tails. We can see `market` factor is not well described by all methodology. On other factors there are some mismatch on tail but roughly the follows the 45 degree line.



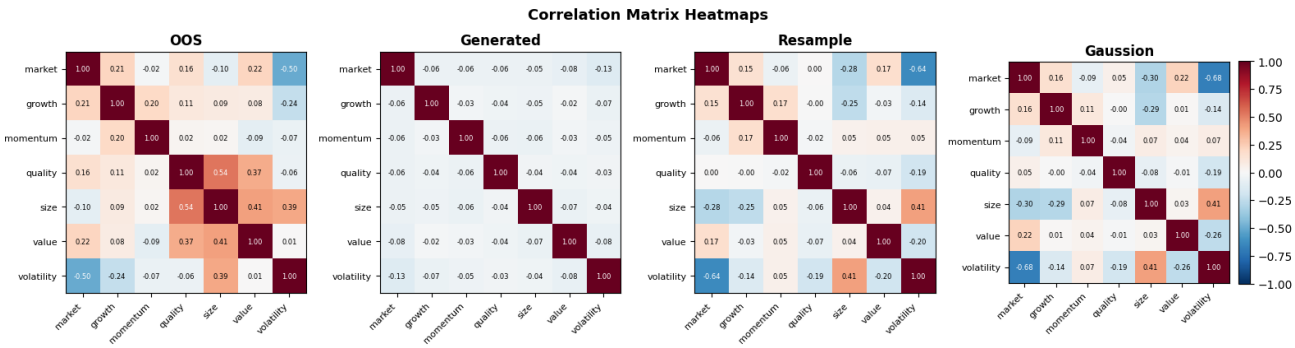
# Kolmogorov–Smirnov test

The Kolmogorov–Smirnov (KS) test is a non-parametric test that compares the empirical distribution functions of two RV. A higher p-value indicates that we cannot reject the null hypothesis that the two RV come from the same distribution. Diffusion model have (4/7) factors better than the resample baseline. And (4/7) factors pass the test if set threshold at 0.05.

```
KS p-value
Factor              Generated       Resample        Gaussion
-------------------------------------------------------------------
market              0.0400          0.0095          0.0000
growth              0.1664          0.1973          0.0031
momentum            0.0387          0.0395          0.0070
quality             0.6017          0.4977          0.0074
size                0.4565          0.5710          0.3447
value               0.0009          0.0002          0.0032
volatility          0.2298          0.0814          0.0006
```

# Factor movement

Correlation matrix heatmaps of the OOS data, generated samples, and baselines. And we use the Fresenius distance between covariance matrices of test data and sampled data to summarize the overall similarity in factor relationships. The heatmaps show that the diffusion samples are very bad in capturing the factor relationship.



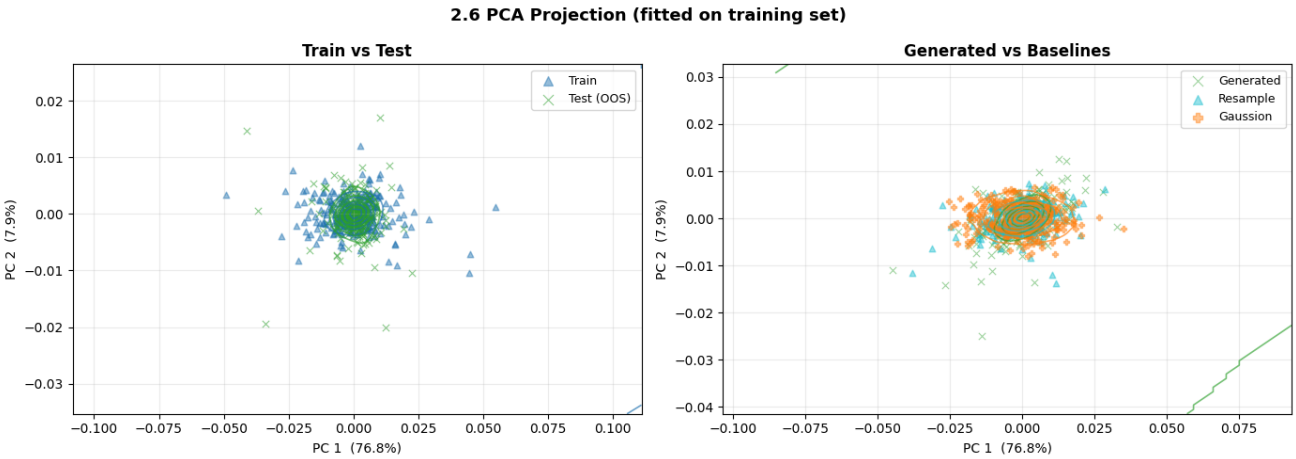

```
Relative Frobenius Distance of Covariance Matrix
  Generated         0.8393
  Resample          0.7911
  Gaussion          0.9316
```

# PCA Projection

We perform PCA on the training set to identify the main axes of variation and project all datasets (test dataset & generated) onto the top 2 PCs for visualization.

PC 1 (76.8% variance, dominated by the `market` factor) captures the primary source of variation. In the left panel, Train and Test overlap well in the central region, but OOS contains several extreme points along PC1 that fall outside the training bulk, which are unseen patterns in the training data.

In the right panel, the three methods show a clear ordering by spread: Gaussian is tightly concentrated at the center, Resample covers a range comparable to training data, and Generated extends furthest along PC1, wider even than the OOS range




```
market        0.942345
volatility   -0.314231
size         -0.096292
value         0.054544
momentum     -0.022634
growth        0.020999
quality       0.006643
dtype: float64
```

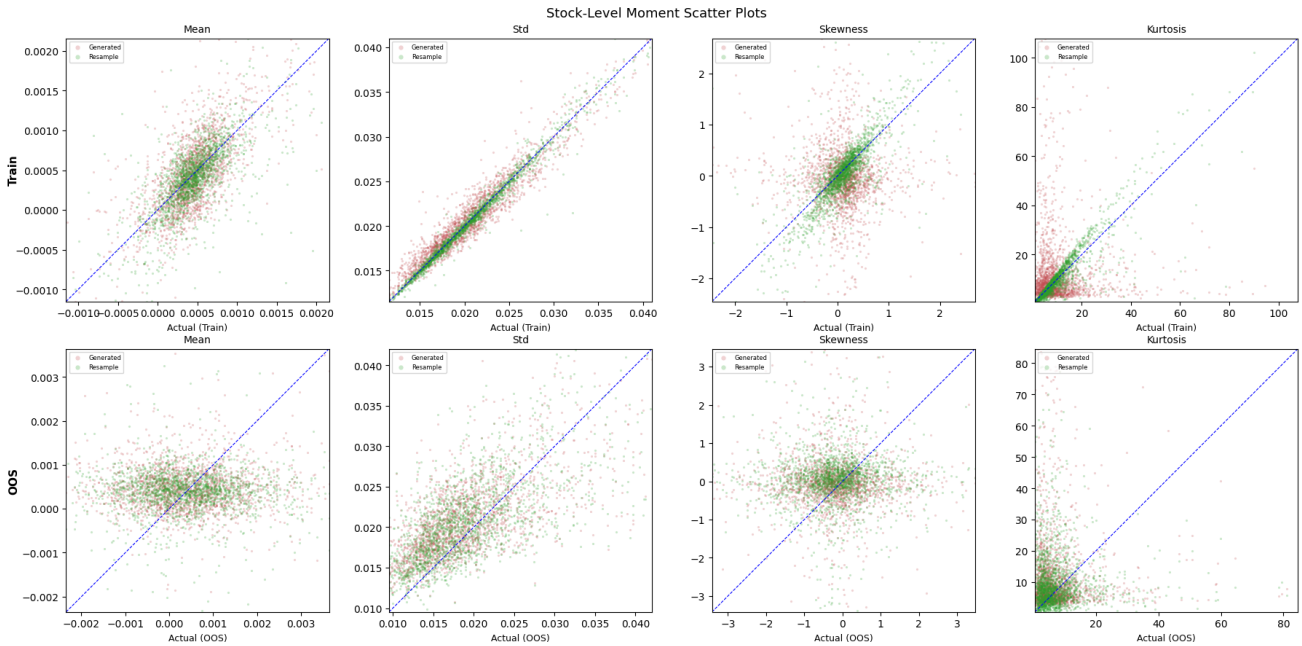
The following are stock comparison, We have 1924 in total. stock returns are reconstructed via the diffusion generated factor returns and the OLS estimated factor loadings. The idiosyncratic part is generated via sampling from t distribution fitted on training residuals.

```
Train returns: (6262, 1924)  NaN fraction: 39.33%
Test returns:  (261, 1924)   NaN fraction: 4.27%
Resample returns: (4096, 1924)  NaN fraction: 39.35%
```

# Moment Comparison (stock level)

We evaluate the moments of stock returns across the common stocks, the first row is sampled against the training set and the second row is sampled against the OOS test set. And we use Pearson correlation to summarize the overall similarity of the moments across stocks.

The diffusion method performs worse than the baseline and fail to capture the higher orders moments even in the training set. And the baseline also fail to adapt to the OOS data on those metrics.

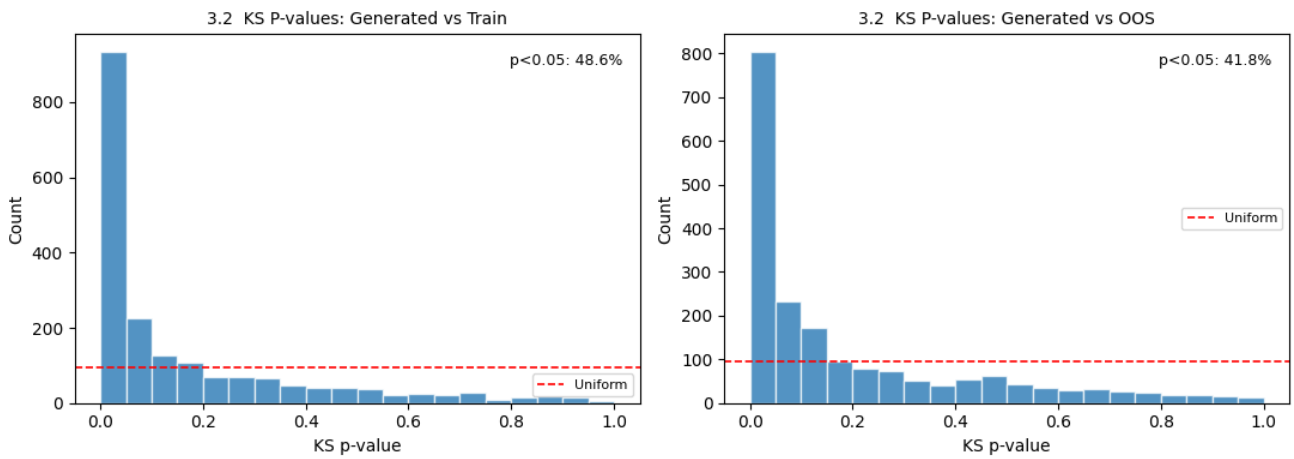



Pearson $R^2$

| | Mean | Std | Skewness | Kurtosis |
|---|---|---|---|---|
| Train Generated | 0.5818 | 0.9235 | 0.0006 | 0.0019 |
| Train Resample | 0.4895 | 0.9828 | 0.7184 | 0.9243 |
| OOS Generated | 0.0001 | 0.4609 | 0.0000 | 0.0004 |
| OOS Resample | 0.0000 | 0.4521 | 0.0011 | 0.0008 |

# Kolmogorov–Smirnov test (stock level)

We perform KS test for train/test set and illustrates the distribution of p-values across stocks. We set p=0.05 as the threshold for passing the test, and count the fraction of stocks that pass the test for both generated and baseline samples against the baseline, and the diffusion model performs worse.



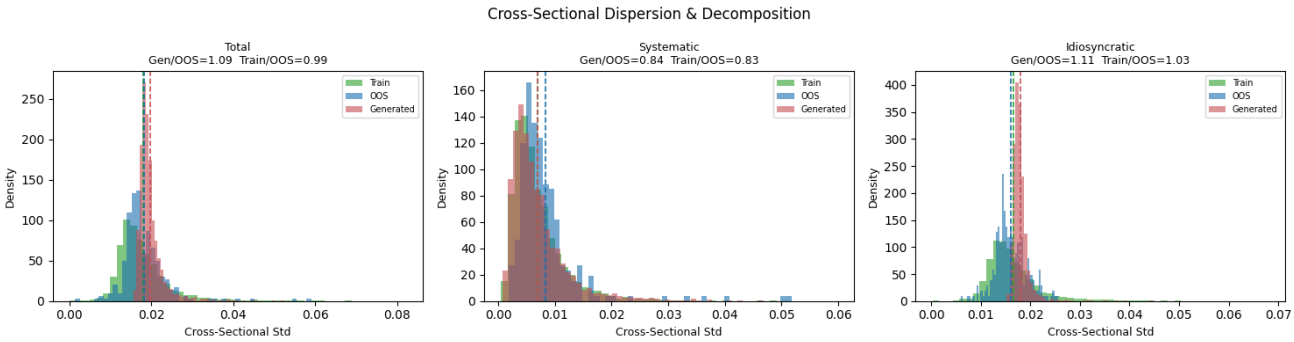

KS p>0.05:

| | Generated | Resample |
|---|---|---|
| vs Train | 51.4% | 100.0% |
| vs OOS | 58.2% | 71.2% |

# Dispersion Comparison

We decompose the returns into systematic (factor-driven) and idiosyncratic components to see how well the model captures each part. And we plot histogram of std on each part and combined, and summarize them via w1 distance.

The results shows that the diffusion model have the same level performance on the systematic level, however the idiosyncratic highly deviate from the OOS. And this deviation dominates the total return level such that diffusion model is worse than the baseline.




```
Wasserstein-1 Distance of Cross-Sectional Std (vs OOS):
Component       W1(Gen, OOS)        W1(Resample, OOS)
-------------------------------------------------------
Total           0.0023              0.0013
Systematic      0.0014              0.0014
Idiosyncratic   0.0022              0.0014
```